MEMBRANES

Module map

- Module 05: Membranes
- Connected lab: lab-06_membranes.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 05
Membranes

1. Phospholipid bilayer structure
2. Selective permeability
3. Passive transport
4. Active transport
5. Osmosis and cell environments

Lab evidence
lab-06_membranes.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

MEMBRANES

Learning objectives

- Describe how phospholipid structure creates a bilayer.
- Predict which molecules cross membranes easily.
- Compare passive and active transport.
- Use concentration gradients to explain movement across membranes.
- Predict osmosis outcomes in hypotonic, isotonic, and hypertonic conditions.

OBJECTIVE 1
Describe how phospholipid structure creates a bilayer.

OBJECTIVE 2
Predict which molecules cross membranes easily.

OBJECTIVE 3
Compare passive and active transport.

OBJECTIVE 4
Use concentration gradients to explain movement across membranes.

OBJECTIVE 5
Predict osmosis outcomes in hypotonic, isotonic, and hypertonic conditions.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

MEMBRANES

Topic sequence

- Phospholipid bilayer structure: The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.
- Selective permeability: Selective permeability controls which materials cross easily and which require help.
- Passive transport: Diffusion, osmosis, and facilitated diffusion move substances down gradients.
- Active transport: Active transport uses energy to move substances against gradients.
- Osmosis and cell environments: Tonicity predicts water movement and cell volume changes.

01

Phospholipid bilayer structure

The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.

02

Selective permeability

Selective permeability controls which materials cross easily and which require help.

03

Passive transport

Diffusion, osmosis, and facilitated diffusion move substances down gradients.

04

Active transport

Active transport uses energy to move substances against gradients.

05

Osmosis and cell environments

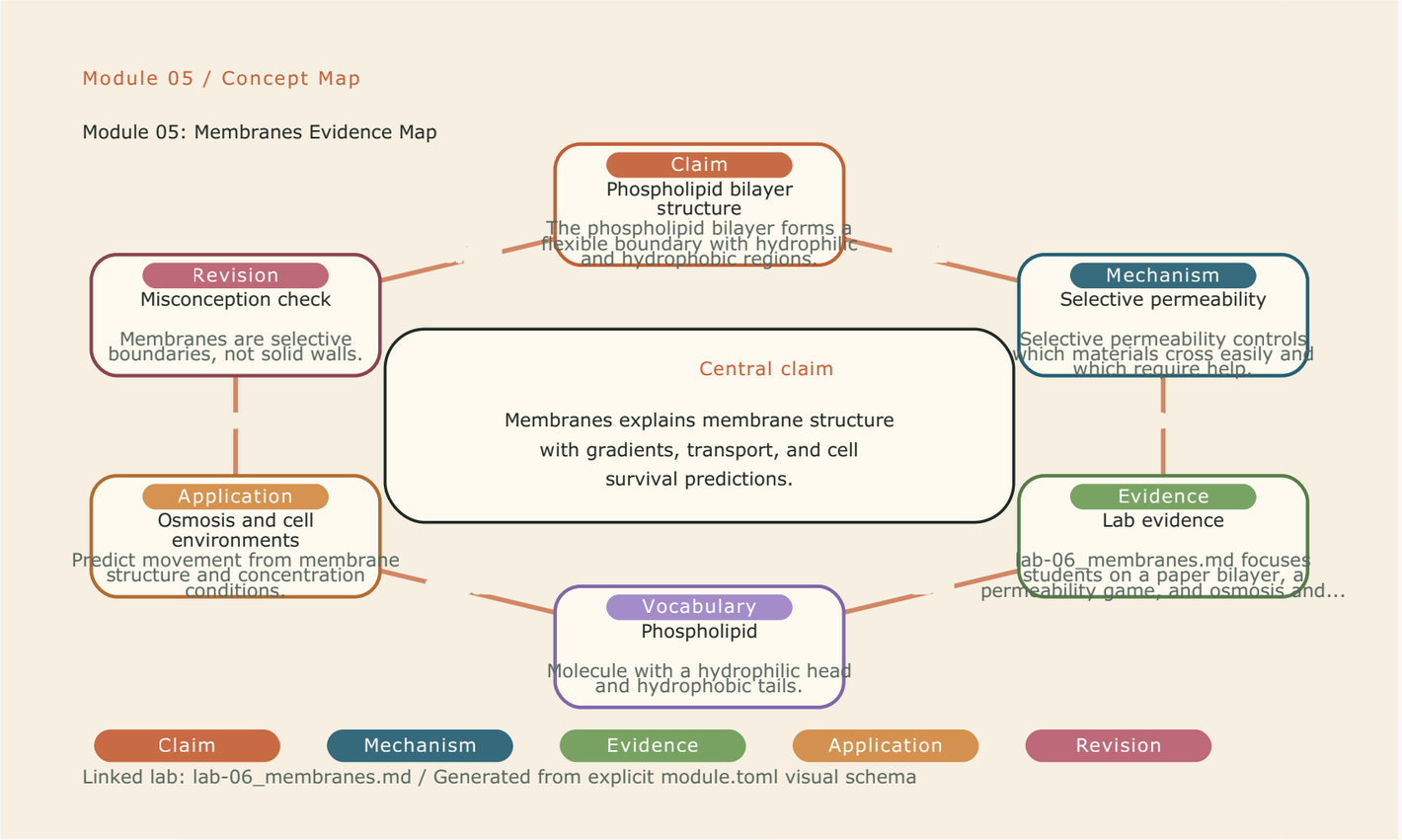
Tonicity predicts water movement and cell volume changes.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

MEMBRANES

Module 05: Membranes Evidence Map

- Module focus: Membranes
- Central claim: Membranes explains membrane structure with gradients, transport, and cell survival predictions.
- Key nodes to connect: Phospholipid bilayer structure, Selective permeability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-06_membranes.md supplies an evidence surface for the map.

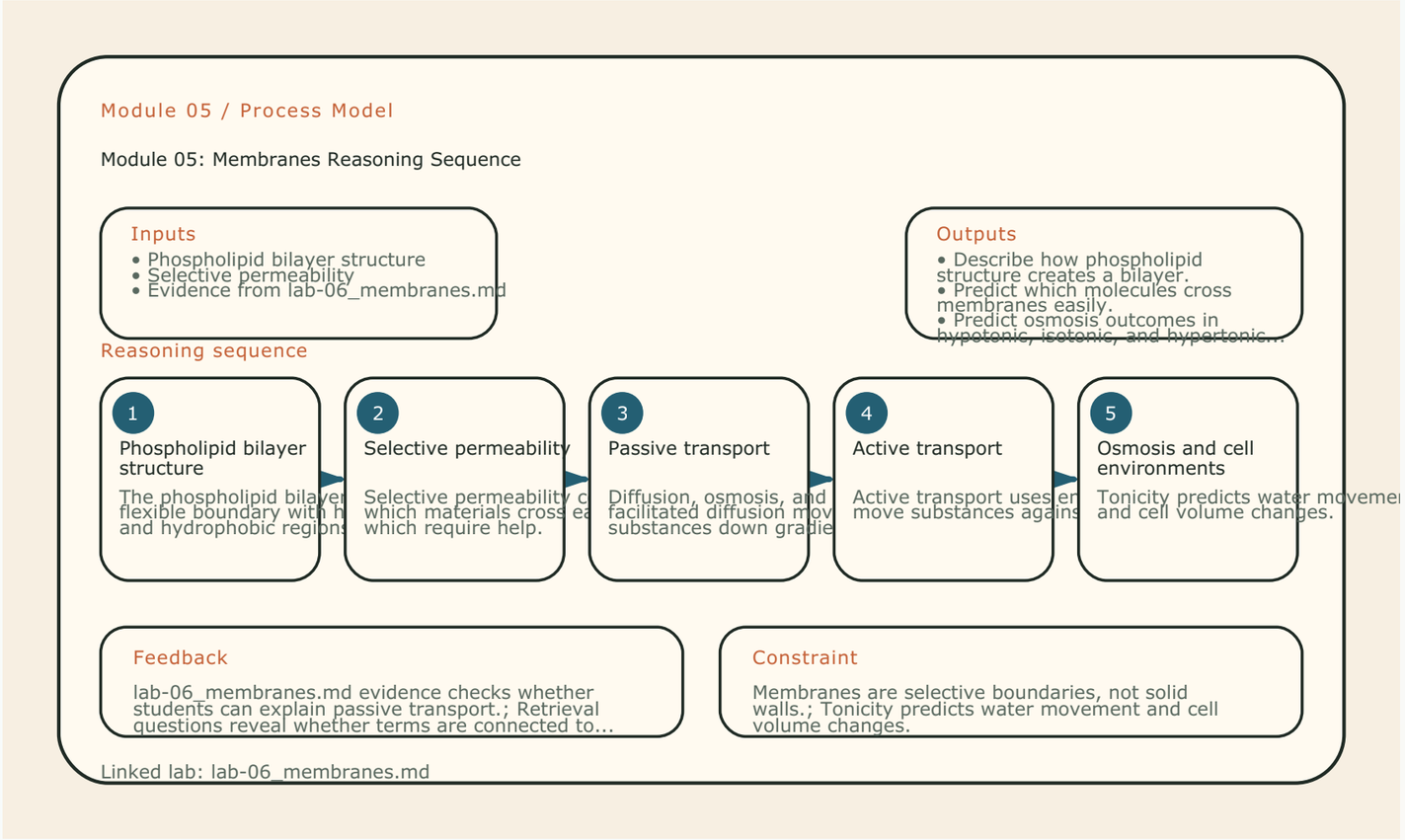


Teaching note: Ask students to explain one edge in their own words before moving on.

MEMBRANES

Module 05: Membranes Reasoning Sequence

- Module focus: Membranes
- Trace the model: Phospholipid bilayer structure -> Selective permeability -> Passive transport -> Active transport
- Inputs become outputs: Phospholipid bilayer structure, Selective permeability -> Describe how phospholipid structure creates a bilayer., Predict which molecules cross membranes easily.
- Feedback check: lab-06_membranes.md evidence checks whether students can explain passive transport.
- Lab connection: lab-06_membranes.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

MEMBRANES

Terms as evidence handles

- Phospholipid: Molecule with a hydrophilic head and hydrophobic tails.
- Bilayer: Double layer of phospholipids forming a membrane.
- Selective permeability: Property of allowing some substances through more easily than others.
- Diffusion: Net movement from high to low concentration.
- Osmosis: Diffusion of water across a selectively permeable membrane.
- Facilitated diffusion: Passive transport through membrane proteins.

PHOSPHOLIPID
Molecule with a hydrophilic head and hydrophobic tails.

BILAYER
Double layer of phospholipids forming a membrane.

SELECTIVE PERMEABILITY
Property of allowing some substances through more easily than others.

DIFFUSION
Net movement from high to low concentration.

OSMOSIS
Diffusion of water across a selectively permeable membrane.

FACILITATED DIFFUSION
Passive transport through membrane proteins.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

MEMBRANES

Lab connection

- Lab file: lab-06_membranes.md
- Lab evidence should test or illustrate: Passive transport
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Phospholipid bilayer structure

EVIDENCE
lab-06_membranes.md

CLAIM
Describe how phospholipid structure creates a bilayer.

Teaching note: Make the lab connection explicit before students encounter the activity.

MEMBRANES

Contrast check

- Do not stop at: The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.
- Stronger explanation: Tonicity predicts water movement and cell volume changes.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.

DEEPER

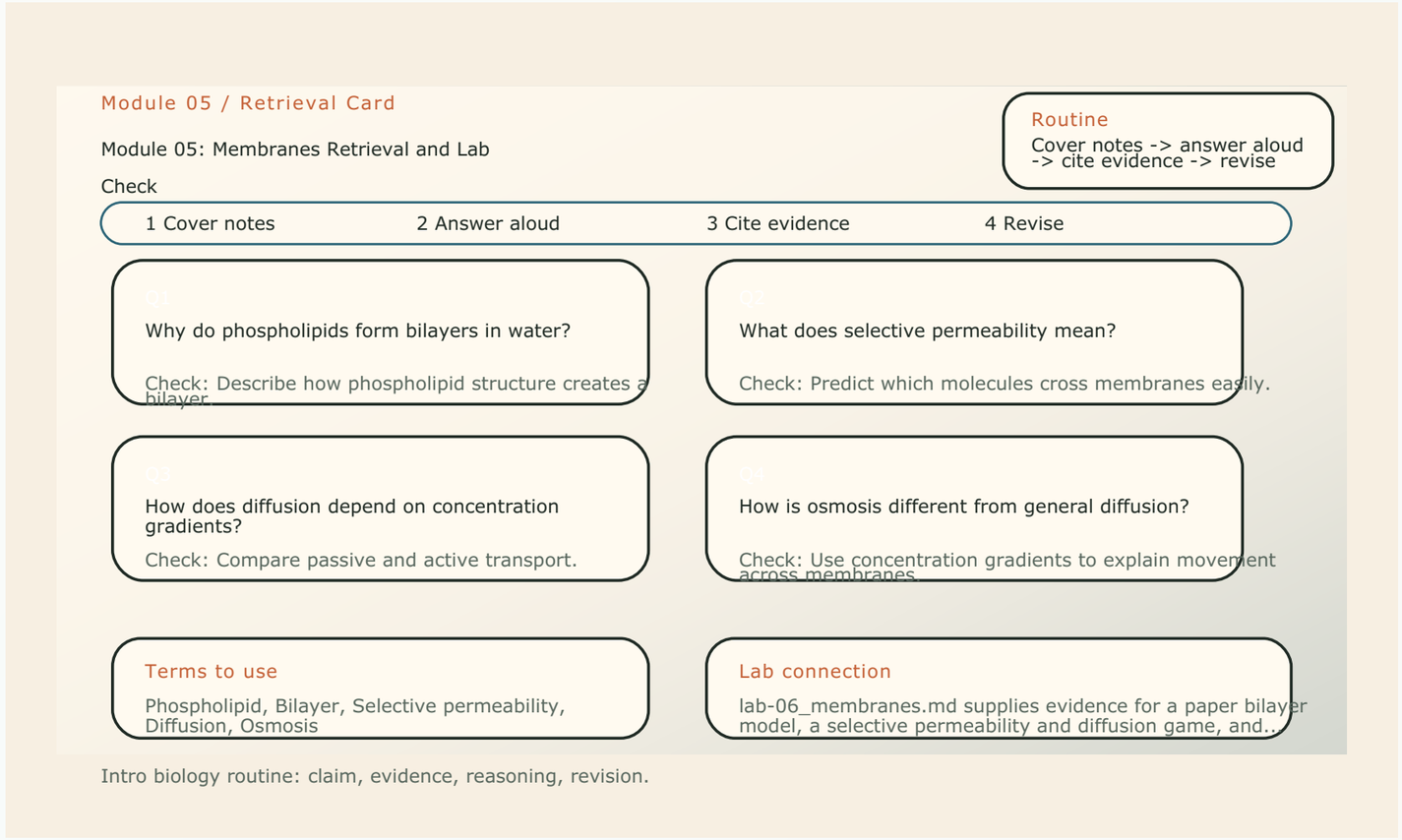
Tonicity predicts water movement and cell volume changes.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

MEMBRANES

Module 05: Membranes Retrieval and Lab Check

- Module focus: Membranes
- Retrieval prompt: Why do phospholipids form bilayers in water?
- Answer check: Describe how phospholipid structure creates a bilayer.
- Required terms: Phospholipid, Bilayer, Selective permeability
- Lab connection: lab-06_membranes.md; lab-06_membranes.md supplies evidence for a paper bilayer model, a selective permeability and diffusion game, and osmosis and tonicity predictions.



Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Why do phospholipids arrange themselves into a bilayer in water?
- 2. A small nonpolar molecule such as oxygen gas crosses the membrane mainly by:
- 3. An animal cell is placed in pure water, a hypotonic solution. What happens?
- 4. Moving ions against their concentration gradient requires:

QUESTION 1

Key: C

Why do phospholipids arrange themselves into a bilayer in water?

QUESTION 2

Key: A

A small nonpolar molecule such as oxygen gas crosses the membrane mainly by:

QUESTION 3

Key: D

An animal cell is placed in pure water, a hypotonic solution. What happens?

QUESTION 4

Key: B

Moving ions against their concentration gradient requires:

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

MEMBRANES

Synthesis and exit ticket

- Exit claim: explain phospholipid bilayer structure using evidence from lab-06_membranes.md.
- Must include: Phospholipid, Bilayer, Selective permeability.
- Revision prompt: what evidence would change your explanation?

CLAIM
Phospholipid bilayer structure

EVIDENCE
lab-06_membranes.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.